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Software Requirements Specification

Project: Decentralized Smart Power Grid

Department of Computer & Electronics Engineering

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1 Introduction

1.1 Purpose

This document specifies the requirements for the Decentralized Smart Power Grid. It is intended for the development team, the instructor, and any stakeholder who needs to understand what the system must do before it is built.

1.2 Scope

The Decentralized Smart Power Grid will allow librarians to manage a book catalogue and members, and allow members to search and reserve books. It replaces the existing manual register and enforces borrowing rules automatically. Out of scope: payment processing and inter-library transfers.

1.3 Definitions, Acronyms, and Abbreviations

SRS	Software Requirements Specification
LMS	Library Management System
ISBN	International Standard Book Number
FR / NFR	Functional / Non-functional Requirement

1.4 Overview of the Document

Summarises the SRS.

2 Objectives

Aim: To elicit the requirements of the chosen system and document them as a Software Requirements Specification using the IEEE 830 template.

Tools used: L^AT_EX for the document.

3 Project Domain Analysis

3.1 Domain Description

A library maintains a collection of books, registers members, and tracks loans. Typical operations: add new books, register members, issue books, return books, and handle reservations. The domain involves two primary actors: Librarian (staff) and Member (patron).

3.2 Existing System Overview

Currently, the library uses a paper-based register. The librarian manually records book issues, due dates, and returns. Searching for a book requires browsing shelves or card catalogues. Reservation is not possible; members must visit physically to check availability.

3.3 Problems in Existing System

The existing library management system is largely manual and relies on paper-based records for maintaining book inventories, member information, and borrowing transactions. This approach presents several challenges:

- **Time-consuming operations:** Recording book issues, returns, and member details manually requires significant time and effort.
- **Difficulty in searching records:** Locating information about a particular book or member from physical registers is slow and inefficient.
- **Human errors:** Manual data entry can lead to mistakes in book records, issue dates, return dates, and member information.
- **Poor tracking of book availability:** Determining whether a book is available, issued, or reserved is difficult without real-time updates.
- **Inefficient management of overdue books:** Tracking due dates and identifying overdue books requires manual checking, which may result in delays and inaccuracies.
- **Risk of data loss:** Physical records can be damaged, misplaced, or lost due to accidents or improper handling.
- **Limited accessibility:** Members must visit the library physically to check book availability or obtain information about library resources.
- **Scalability issues:** Managing a large collection of books and a growing number of members becomes increasingly difficult using manual methods.

3.4 Proposed System

The proposed LMS is a web application with a relational database. It automates catalogue management, enforces borrowing rules, provides instant search, and allows online reservations. It reduces manual errors and improves accessibility.

4 Software Process Model

4.1 Waterfall Model

The fall model (classic life cycle) is chosen: Requirements → Design → Implementation → Testing → Deployment → Maintenance.

4.2 Process Model Diagram

Include necessary diagrams

4.3 Phases of the Model

1. **Requirements analysis:** This SRS is the output.
2. **System design:** Architecture, database schema, component design.
3. **Implementation:** Coding the web application.
4. **Testing:** Unit, integration, and system testing against the SRS.
5. **Deployment:** Install on library server.
6. **Maintenance:** Bug fixes and minor enhancements.

4.4 Justification

- Requirements of LMS are well-understood and stable (typical library operations).
- The project is small to medium in size, with a clear sequence of phases.
- The team has previous experience with similar academic projects.
- Milestones (SRS, design document, tested system) are easy to evaluate.

5 Overall Description

5.1 Product Perspective

The Decentralized Smart Power Grid is a standalone web application backed by a relational database. It interacts with library staff and members through a browser.

5.2 Product Functions

- Manage the book catalogue (add / edit / remove books).
- Search the catalogue by title or author.
- Issue and return books, enforcing borrowing limits and due dates.
- Reserve unavailable books.
- Manage member records.

5.3 User Characteristics

- **Librarian:** trained staff member; performs all catalogue and loan operations.
- **Member:** general user; searches and reserves books.
- **System Administrator:** Maintains servers, backups, and user accounts.

5.4 Key Entities

- **Book:** ISBN, title, author, publisher, year, quantity, available copies.
- **Member:** Member ID, name, contact, maximum allowed books.
- **Transaction:** Issue date, due date, return date, book copy, member.
- **Reservation:** Book ID, member ID, reservation date, status.

5.5 Constraints

- Must run on the institution's existing server infrastructure.
- Passwords must be stored only as salted hashes.

5.6 Assumptions and Dependencies

- Members are registered before they can reserve books.
- A network connection to the database server is available.

6 Specific Requirements

6.1 External Interface Requirements

6.1.1 User Interfaces

A browser-based interface with a catalogue search page, a librarian dashboard, and a member account page.

6.1.2 Hardware Interfaces

Standard PC / server hardware; optionally a barcode scanner for ISBNs.

6.1.3 Software Interfaces

A relational database (e.g., PostgreSQL) accessed by the application.

6.1.4 Communication Interfaces

HTTPS for all client-server traffic.

6.2 Functional Requirements

ID	Functional Requirement	Priority
FR-1	The system shall allow a librarian to add, edit, and remove books in the catalogue.	High
FR-2	The system shall allow a member to search the catalogue by title or author.	High
FR-3	The system shall allow a librarian to issue an available book to a registered member.	High
FR-4	The system shall reject an issue when the member already holds the maximum allowed books.	High
FR-5	The system shall set a due date 14 days from the issue date.	Medium
FR-6	The system shall allow a member to reserve a book that is currently unavailable.	Low

6.3 Non-functional Requirements

ID	Non-functional Requirement	Category
NFR-1	A catalogue search shall return results within 2 seconds for a 10,000-book catalogue.	Performance
NFR-2	Passwords shall be stored only as salted hashes.	Security
NFR-3	The system shall be usable on a 1366×768 screen without horizontal scrolling.	Usability
NFR-4	The system shall be available 99% of the time during library opening hours.	Reliability

6.4 Other Requirements

If any.

7 Conclusion

In this lab the team elicited and documented the requirements of the Decentralized Smart Power Grid using the IEEE 830 template. The functional and non-functional requirements were each given unique identifiers so they can be traced into the design and test phases in later labs.

8 References

1. IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications.
2. more reference if any.